

Dolutegravir Resistance Mutation R263K Cannot Coexist in Combination with Many Classical Integrase Inhibitor Resistance Substitutions

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The new integrase strand transfer inhibitor (INSTI) dolutegravir (DTG) displays limited cross-resistance with older drugs of this class and selects for the R263K substitution in treatment-experienced patients. We performed tissue culture selections with DTG, using viruses resistant to older INSTIs and infectivity and resistance assays, and showed that the presence of the E92Q or N155H substitution was compatible with the emergence of R263K, whereas the G140S Q148R, E92Q N155H, G140S, Y143R, and Q148R substitutions were not.

Although considerable progress has been made in HIV therapy, drug resistance has emerged for every available drug class, threatening long-term care and the potential for a cure (1). The first two integrase strand transfer inhibitors (INSTIs), raltegravir (RAL) and elvitegravir (EVG), can be compromised by primary substitutions that generate resistance at the expense of fitness (2), the majority of which occur in the catalytic core domain of the integrase protein (IN). Although the newer INSTI dolutegravir (DTG) has not yielded resistance when used in initial therapy (3), our group selected an R263K substitution in HIV-1 IN in tissue culture with DTG (4). We showed that R263K, located in the C-terminal portion of the protein, confers moderate resistance to DTG and EVG, while also decreasing the biochemical activity of IN by ~30% (5). The R263K substitution was also found in two of four INSTI-naïve treatment-experienced participants who failed DTG therapy in the SAILING clinical trial (of 354 patients) but otherwise continued to perform well (6). This suggests that R263K is a signature substitution for DTG.

In contrast, 7/51 individuals who previously failed RAL-based regimens and who possessed primary resistance mutations subsequently failed DTG-based therapy in the VIKING trial without R263K (7), suggesting that the substitutions that confer resistance to RAL or EVG might be incompatible with R263K.

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To test this hypothesis, biochemical analyses of the strand transfer efficiencies of integrase proteins containing the E92Q, Y143R, Q148R, and N155H RAL or EVG resistance substitutions, both alone and in combination with R263K, were performed (Fig. 1). Site-directed mutagenesis was performed as described previously (4, 5) to introduce E92Q, Y143R, Q148R, and N155H into the wild-type (pET15b_{wt}) and mutant (pET15b_{R263K}) integrase subtype B expression vectors (primer sequences available upon request). Recombinant integrase proteins were expressed and purified and strand transfer assays were performed as published (4, 5). Table 1 summarizes the enzyme kinetics values K_m and V_{max} for each protein tested.

Every single mutation had a significant negative impact on strand transfer activity, and this effect was exacerbated when these substitutions were combined with R263K. Among single mutant proteins, the Q148R- and N155H-containing integrases were the most impacted with regard to K_m , with fold changes of 2.49 and 2.72 from the wild type (WT), respectively. Combinations of E92Q and R263K or Q148R and R263K resulted in the lowest levels of strand transfer activity (Fig. 1 and Table 1).

We then measured the effects of these mutations in isolation or with R263K on the relative infectiousness of NL4.3 virus as described previously (5). HIV-1 infectivity was measured through the infection of 30,000 TZM-bl cells per well using serial 1:4 dilutions of the NL4.3 variants. After 48 h, cells were lysed, and luciferase production was measured using the Promega luciferase assay system (Promega, Madison, WI). Each single substitution had a negative impact on infectiousness. The NL4.3_{IN(Y143R R263K)} and NL4.3_{IN(Q148R R263K)} viruses showed the greatest declines with increases in relative 50% effective concentration (EC_{50}) of 17.9- and 29.6-fold compared to the wild type, respectively. The NL4.3_{IN(N155H R263K)} virus had a significantly lower EC_{50} than did the NL4.3_{IN(R263K)} single mutant (fold change of 1.2 for N155H R263K compared to 2.7 for R263K alone), suggesting that the addition of N155H partially restored the deficit in infectivity observed with R263K.

We next measured HIV-1 susceptibility to DTG, RAL, and

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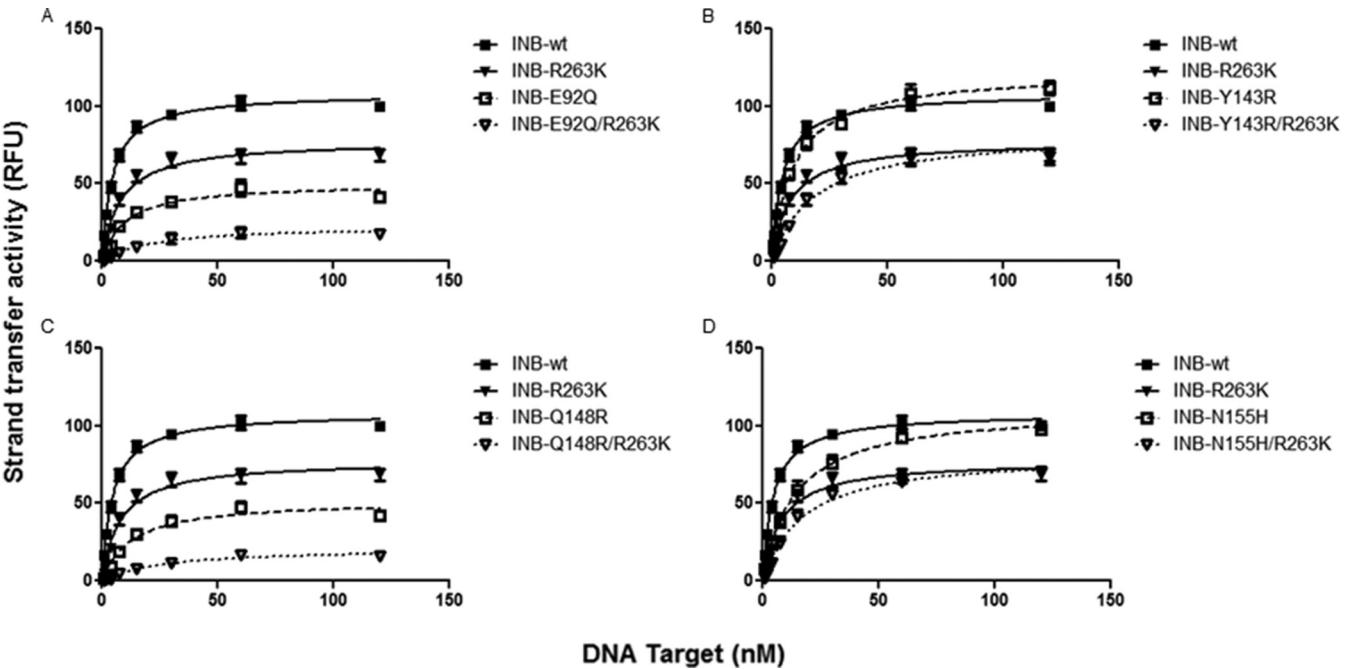


FIG 1 Relative strand transfer efficiencies of mutant integrase proteins when concentrations of biotinylated target DNA were varied. Wild-type and R263K integrase proteins were included as internal controls in each panel. (A) E92Q ± R263K; (B) Y143R ± R263K; (C) Q148R ± R263K; (D) N155H ± R263K. Error bars show the standard error of the mean (SEM). The addition of R263K to each single-resistance mutation consistently led to a decrease in strand transfer activity.

EVG in tissue culture through the determination of the 50% inhibitory concentration (IC_{50}) (Table 2). HIV susceptibilities to INSTIs were measured through infection of 30,000 TZM-bl cells using 200,000 reverse transcriptase (RT) units per well of each virus in the presence of 1:4 serial dilutions of drugs. Levels of infection were measured as described above. The NL4.3_{IN(R263K)} virus displayed a 2-fold increase in resistance against DTG compared to the wild type, as previously observed (4, 5, 8, 9). The impairment of replication seen with the NL4.3_{IN(Y143R R263K)} and NL4.3_{IN(Q148R R263K)} viruses impeded data collection. The addition of R263K to NL4.3_{IN(E92Q)} signif-

icantly increased the IC_{50} for DTG to 24.9 nM compared to 18.2 nM for the WT (1.4-fold), whereas the IC_{50} s were 13.2 nM and 36.3 nM for NL4.3_{IN(E92Q)} and NL4.3_{IN(R263K)}, respectively. Importantly, the addition of N155H to R263K, which partially restored the defect in infectiousness associated with the latter mutation, also increased resistance against DTG (from 2.0- to 3.3-fold relative to the WT). The addition of R263K to the classical INSTI resistance substitutions did not have a major effect on HIV-1 susceptibility to RAL. The NL4.3_{IN(E92Q)} virus was highly resistant against EVG, and resistance levels were unaffected by the addition of R263K. In

TABLE 1 Effect of classical INSTI resistance substitutions on cell-free strand transfer activity and NL4.3 infectivity in tissue culture

Genotype	Result from ^a :					NL4.3-infected TZM-bl cell assay	
	Cell-free strand transfer assays					Relative EC_{50}	95% CI
	K_m (nM)	95% CI	FC	V_{max}			
WT	5.07	4.30–5.85	1.00	100.00		1.00	0.67–1.48
E92Q	9.67	4.64–14.69	1.91*	46.27		1.46†	1.08–1.96
Y143R	9.39	7.81–10.98	1.85*	112.88		1.28†	0.97–1.69
Q148R	12.60	5.17–20.03	2.49*	48.07		6.06*†	4.84–7.59
N155H	13.78	12.07–15.50	2.72*	102.87		1.27†	0.77–2.08
R263K	9.94	7.48–12.39	1.96*	68.72		2.66*	1.90–3.73
E92Q R263K	21.23	9.68–32.78	4.19*†	21.08		3.78*	2.72–5.26
Y143R R263K	18.03	10.63–25.43	3.56*†	76.69		17.98*†	11.61–27.60
Q148R R263K	26.64	8.83–44.46	5.26*†	19.70		29.61*†	8.53–102.83
N155H R263K	16.99	11.50–22.47	3.35*†	76.14		1.19†	0.86–1.64

^a Shown are the effects of primary integrase strand transfer inhibitor (INSTI) resistance mutations ± R263K on the K_m and V_{max} (relative to the WT) in cell-free assays, the fold change (FC) for K_m , and the EC_{50} (relative to WT) of NL4.3 virus in TZM-bl cells. The 95% confidence intervals (95% CI) are reported. *, $P < 0.05$ for Student's t test comparing each value (K_m or EC_{50}) to the WT value; †, $P < 0.05$ for Student's t test comparing each value (K_m or EC_{50}) to the R263K mutant value.

TABLE 2 Determination of IC₅₀ values for wild-type and mutant NL4.3 viruses for DTG, RAL, and EVG

Genotype	Result for ^a :								
	DTG			RAL			EVG		
	IC ₅₀ (nM)	95% CI	FC	IC ₅₀ (nM)	95% CI	FC	IC ₅₀ (nM)	95% CI	FC
WT	18.17	15.04–21.93	1.00	114.50	69.69–188.1	1.00	136.9	51.81–361.7	1.00
E92Q	13.20	9.52–18.28	0.73†	325.20	124.8–847.2	2.84†	>5,000		>50*†
Y143R	19.65	15.41–25.06	1.08†	1,173.00	321.6–4,276.0	10.24*†	282	39.27–2,022	2.06†
Q148R	2.33	1.54–3.53	0.13*†	>5,000		>50*†	>5,000		>50*†
N155H	1.81	1.31–2.50	0.10*†	695.80	566.0–855.4	6.08*†	646	387.9–1,076	4.72*†
R263K	36.28	35.24–37.36	2.00*	55.45	32.99–93.19	0.48	2,539	1,819–3,544	18.55*
E92Q R263K	24.86	22.92–26.96	1.37*†	277.60	134.8–571.7	2.42†	>5,000		>50*†
N155H R263K	60.28	29.22–124.4	3.32*	421.90	293.1–607.3	3.68*†	889	439.7–1799	6.49*†
Y143R R263K		No data			No data			No data	
Q148R R263K		No data			No data			No data	

^a Shown are the effects of primary integrase strand transfer inhibitor (INSTI) resistance mutations ± R263K on IC₅₀s in TZM-bl cells for dolutegravir (DTG), raltegravir (RAL), and elvitegravir (EVG). The 95% confidence intervals (95% CI) and fold change (FC) are reported. *, $P < 0.05$ for Student's t test comparing each IC₅₀ to the WT value; †, $P < 0.05$ for Student's t test comparing each IC₅₀ to the R263K mutant value.

contrast, the addition of R263K to N155H increased resistance to EVG by 1.4-fold.

Finally, we investigated whether RAL- or EVG-resistant viruses would permit the emergence of the R263K mutation during tissue culture selections in cord blood mononuclear cells (CBMCs) infected with various viruses in the presence of DTG (Table 3), performed as described previously (10, 11). The NL4.3_{wt}, NL4.3_{IN(E92Q)}, and NL4.3_{IN(N155H)} viruses were all able to select for R263K. Although none of the other combinations of substitutions were compatible with the emergence of R263K, the G140S Q148R mutant virus was able to select for H51Y, a secondary resistance mutation that has been associated with R263K in DTG resistance (5). As previously shown, the NL4.3 virus was unable to select for a compensatory mutation that could restore the decrease in fitness conferred by R263K (5, 9–11), suggesting that E92Q or N155H must be present first if R263K is to subsequently arise.

These results are consistent with recent clinical observations of the N155H resistance pathway in treatment-experienced patients failing therapy with RAL and DTG (12–14). Our results also explain the absence of the R263K mutation in participants in the VIKING clinical trial who failed DTG after having previously

failed either RAL or EVG, as substitutions at positions 140 and 148 were common in that trial (7).

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TABLE 3 Emerging mutations in integrase detected by genotyping viruses containing INSTI resistance-associated mutations that were grown under DTG pressure for 30 weeks^a

Starting genotype	Drug concn (μM)	New mutation(s) selected
WT	0.03	R263K
E92Q	0.05	R263K
N155H	0.10	R263K
E92Q N155H	0.05	None
Q148R	0.25	None ^b
G140S ^c	0.50	V13I, V54I, Q148R
G140S Q148R	0.50	K14R, H51Y/H, V54I

^a Shown are changes in the integrase-coding sequence from HIV-1 subtype B NL4.3 clonal viruses during a 30-week dolutegravir (DTG) selection. The highest drug concentrations reached with each virus are noted.

^b Reversion to Q148.

^c G140S is a common secondary mutation to Q148R.

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